

Traffic Engineering

Using the whole network without losing the ability to recover

Routing capstone



Why traffic engineering exists

Shortest paths answer reachability.

Shortest paths may not be the best way to route may.

Shortest-path routing

Find a valid next hop
Adapt after failures
Run without a global controller



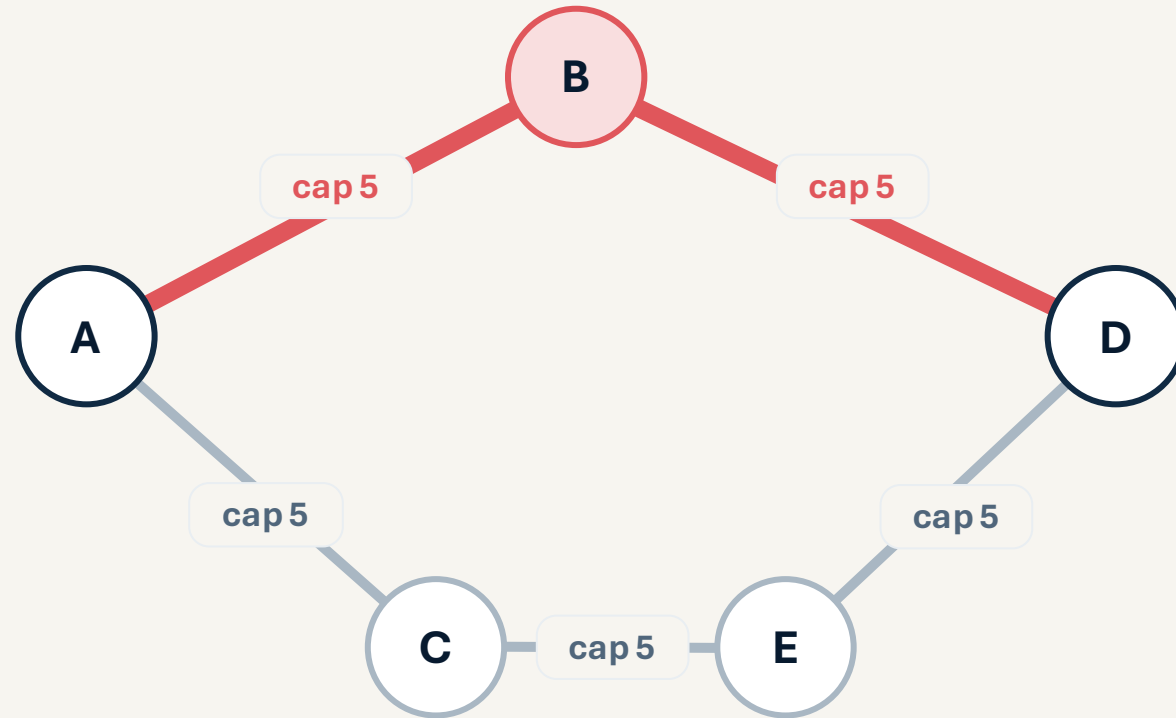
Traffic engineering

Place large demands across paths
Avoid hot links
Honor priorities when capacity is tight

We will not be discussing that

Question: what if two large flows choose the same short link while another path sits idle?

A shortest-path hotspot



Demand A to D

8 units

Shortest path

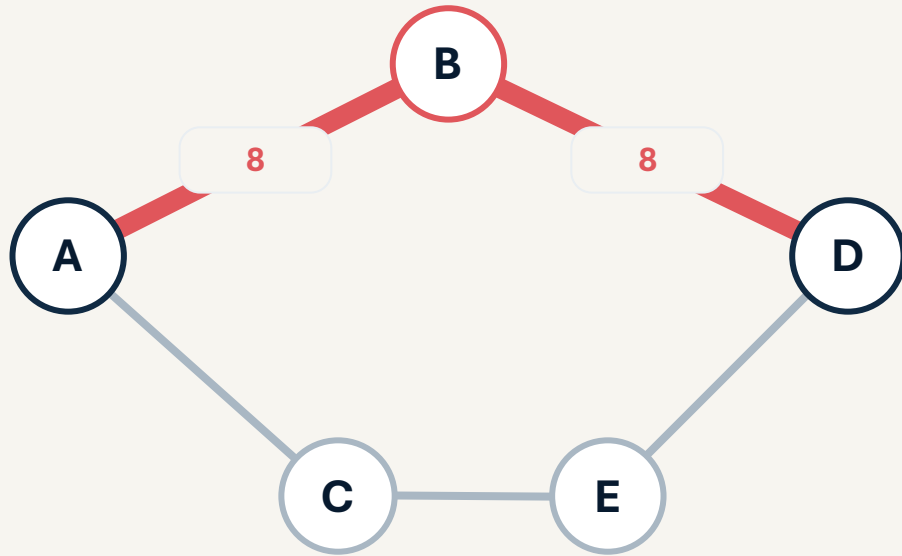
A-B-D

8 offered to a link that carries 5

A longer path has unused capacity.

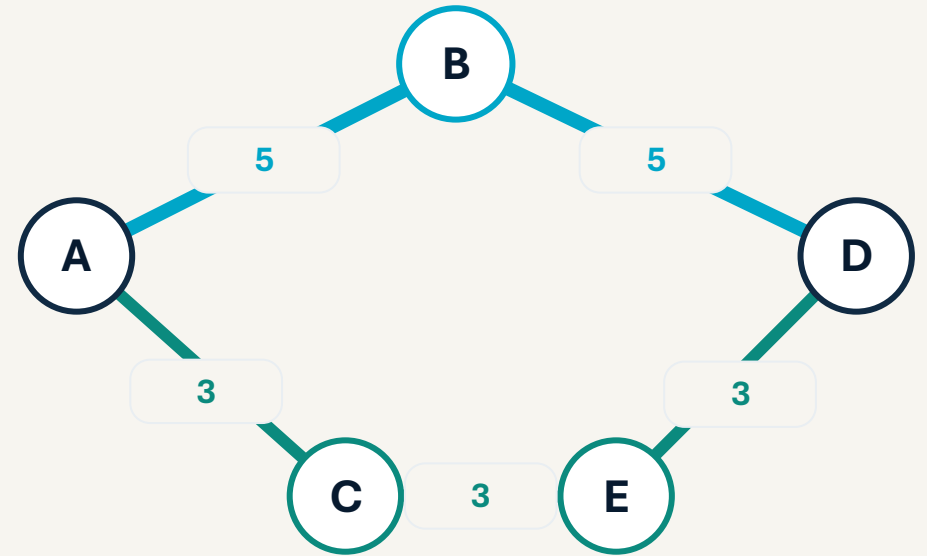
Traffic split across two paths

Shortest path



offered load 8 / capacity 5

Optimal solution



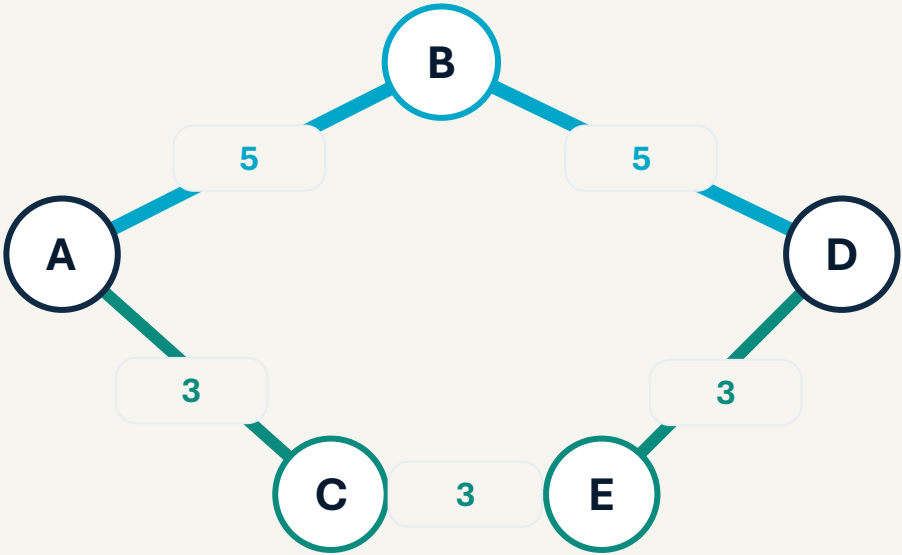
top: 5 bottom: 3 $U = 1$

The LP does not care that the second route has one more hop.

How do we split traffic in IP?

Prefix	Next hop
134.0.0.0/9	B
134.128.0.0/9	C

LP solution



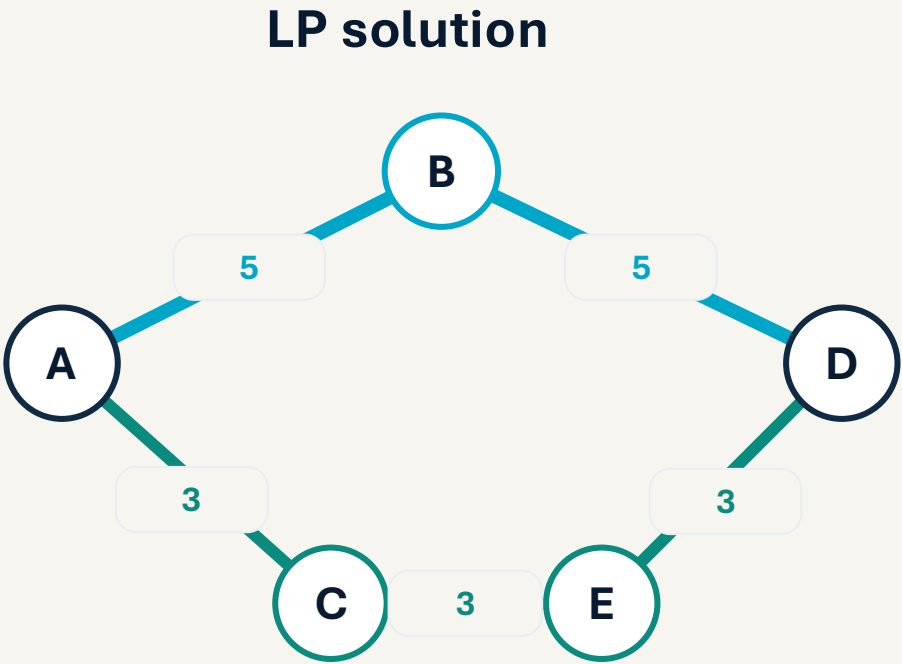
Destination prefixes:
134.0.0.0/8

Does this implement our
desired solution?

top: 5 bottom: 3 U = 1

How do we split traffic in IP?

Prefix	Next hop
134.0.0.0/11	B
134.32.0.0/11	B
134.64.0.0/11	B
134.96.0.0/11	B
134.128.0.0/11	B
134.150.0.0/11	C
134.182.0.0/11	C
134.214.0.0/11	C

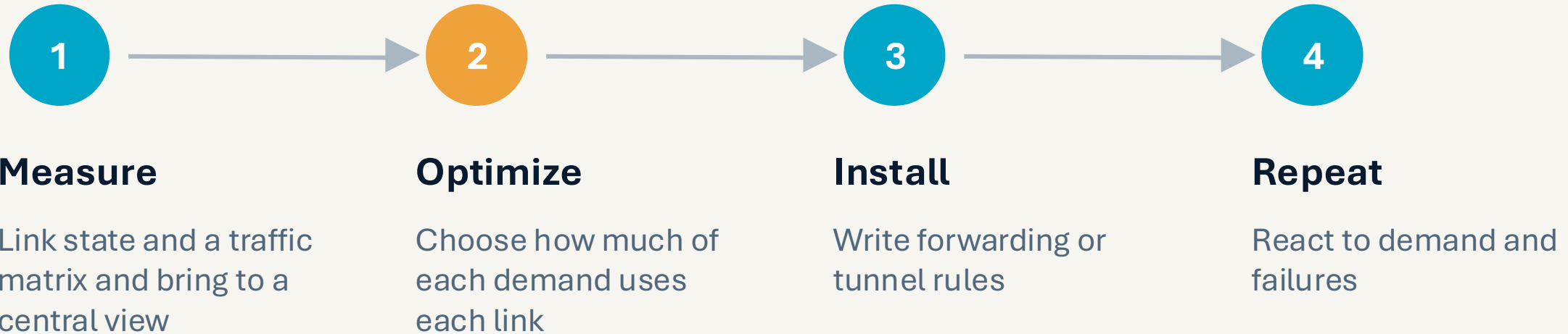


Destination prefixes:
134.0.0.0/8

top: 5 bottom: 3 U = 1

We get a clear 5:3 ratio
Q: Can we compress this table?

The traffic-engineering control loop



Centralized design buys a global view. It also creates a dependency on fresh measurements and controller reachability.

The SDN control stack

MAY RUN ELSEWHERE

Network controller

software on a standard CPU

control messages



INSIDE THE SWITCH

Switch controller / agent

software on the switch CPU

installs forwarding entries ↓

DATAPATH

fast packet-processing hardware

Two pieces of control software

The network controller can be off the switch. The switch controller runs locally.

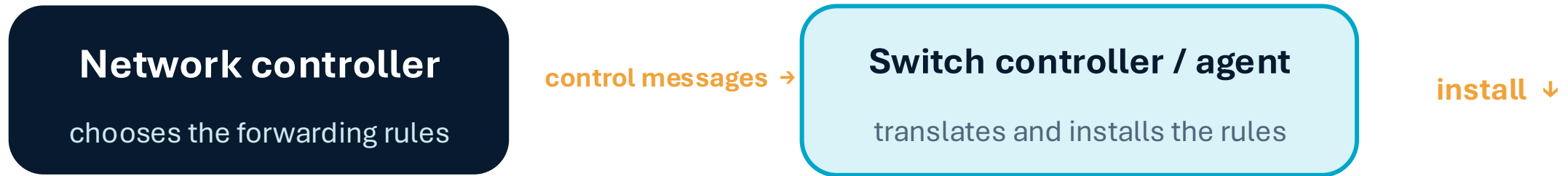
One fast forwarding engine

The datapath applies the installed rules to packets.

Control software decides. Datapath forwards.

Control path and packet path

CONTROL PATH



PACKET PATH



Ordinary packets stay in the datapath.

The controller changes rules. It does not process every packet.

A basic multicommodity-flow LP

One demand is one commodity.

Network	directed graph $G = (V, E)$
Capacity	$c[e]$ on each link e
Demand k	source $s[k]$, destination $t[k]$, volume $d[k]$
Flow variable	$f[k, e] \geq 0$ on every demand-link pair
Utilization	$U \geq 0$, the largest link utilization

Objective

minimize

U

spread traffic so the
busiest link is as quiet as
possible

The LP has two kinds of constraints

1 Flow conservation

$$\text{out}(v,k) - \text{in}(v,k) = \begin{cases} d[k] & \text{at the source} \\ -d[k] & \text{at the destination} \\ 0 & \text{everywhere else} \end{cases}$$

Traffic cannot appear or disappear inside the network.

2 Link capacity

$$\text{For every link } e: \quad \sum_k f[k,e] \leq U \times c[e]$$

All commodities share the same physical link.

The basic LP is expensive

flow variables $\approx$ demands $\times$ links

$$K \times |E|$$

- Every demand gets a variable on every directed link.
- A new traffic matrix or link failure can trigger a network-wide solve.
- The fractional answer must become a finite set of forwarding rules.
- Measurements, computation, and installation all take time.

Useful as a clean model. Inefficient as a literal real-time implementation.

Google's network combines two control styles

Centralized TE overlay

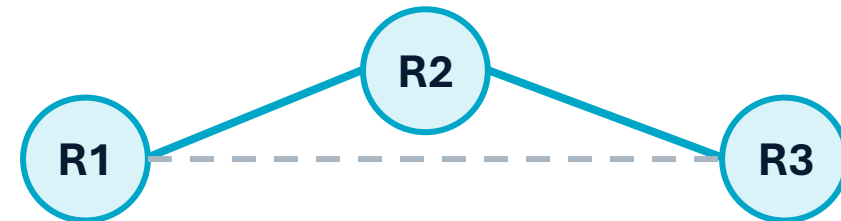
Global demand view
Multipath tunnels
Application priorities

TE service

computes and installs higher-priority forwarding entries

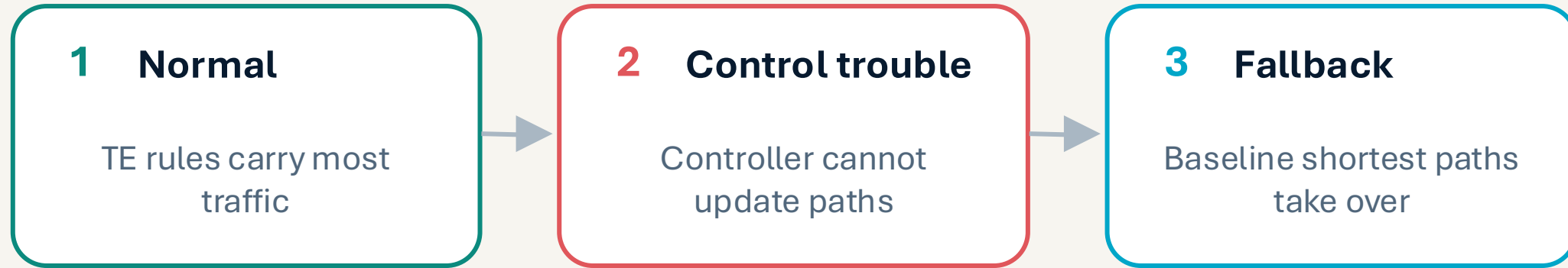
Baseline distributed routing

BGP / IS-IS computes ordinary shortest paths
Lower-priority rules stay installed



Overlay wins while healthy. Baseline remains the safety net.

The shortest-path safety net



A small virtual subnetwork can reserve shortest-path reachability for:

controller messages

routing updates

repair traffic

A controller outage does not have to become a forwarding outage.

Backup capacity can come from another network

Before a failure

- Contract for transit, a leased circuit, or reserved capacity.
- Build the physical interconnection in advance.
- Preconfigure authentication, prefixes, and routing policy.

During a failure

- Activate the backup route.
- Carry the control plane first, then the most important data.

Hypothetical conversation

Google

If my backbone is impaired, carry this narrow control slice.

Verizon

The standby link and policy are already in place.

Failure-time capacity requires pre-failure business and technical work.

One network, two levels of intelligence

Distributed routing

Keeps reachability alive
Converges after failures
Provides the fallback paths

Central traffic engineering

Sees the whole traffic matrix
Splits demand across paths
Uses capacity according to policy

Failure question: which traffic gets bandwidth first?

control and repair · interactive service · bulk transfer